PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

JOINT INSPIRING EXAMINATIONS FOR PRIVATE SECONDARY SCHOOLS

(JIEPSS)

FORM THREE JOINT TERMINAL EXAMINATION

041 BASIC MATHEMATICS

TME: 3 HOURS **MAY 2025**

INSTRUCTIONS

1. This paper consists of section **A** and **B** with total of **Fourteen (14)** questions.
2. Answer **all** questions in both sections
3. Each question in section A carries **six (6)** marks while each question in section B carries **ten (10)** marks.
4. All writings must be in blue or black ink, except drawing which should be in pencil
5. Mathematical tables and non-programmable calculators may be used.
6. All communication devices, programmable calculators and any unauthorized materials are not allowed in examination room.
7. Write your **Examination number** on every page of your answer sheets.

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SECTION A: (60 Marks)

Answer **all** questions in this section

1. (a) A man buys equal numbers of chairs, tables, and shelves. He packs them into identical groups. He finds that the maximum number of groups is 12 and each group has 18 total items. How many chairs, tables, and shelves did he buy?

   (b) Three schools hold math contests every 60, 72, and 108 days. If the schools receive 540 pens, 630 papers, and 720 rulers, what is the greatest number of identical kits they can prepare?

2. (a) Write $\frac{1}{2-\sqrt{3}} - \frac{2}{4+\sqrt{2}}$ in the form of $\frac{a}{b\sqrt{c}-d}$.

   (b) Given that $\log 2 = a$, $\log 3 = b$ and $c = \log 2\frac{1}{4}$. Write a interms of b and c

3. A is a set which contains all two-digit numbers divisible by 6 and B is a set which contains all two digit numbers divisible by 9. If A and B are both integers. Find the following

   (a) $A \cap B$

   (b) The number of elements in $A \cup B$

   (c) A but not B

   (d) Shade $(A \cap B') \cup (A \cup B)$ in a Venn diagram

4. (a) René Descartes, the founder of coordinate geometry used the abscissa and ordinate to locate points. Plot point P whose abscissa is 3 and ordinate is -2, and point Q whose abscissa is -4 and ordinate is 5. Hence state in which quadrants do the points lie?

   (b) If $(2,3)$, $(m,5)$ and $(-6,n)$ are colinear. Find the equation of line through $A(1,m)$ and $B(n,5)$ in the form of $\frac{x}{a} + \frac{y}{b} = 1$.

5. (a) A piece of wire which is $56cm$ long is bent to form a rectangular area of $171cm^2$. Find the dimensions of a rectangle.

   (b) A vertical flagpole casts a shadow of 4.5 meters. At the same time, a 1.5-meter tall stick casts a shadow of 0.9 meters. Are the triangles formed by the flagpole and its shadow and the stick and its shadow similar? Give a reason.

6. (a) The number of workers w needed to repair a road is inversely proportion to the square of time taken t.

      (i) Write an expression to represent the above information

      (ii) If w is increased by 12%. Find the percentage change in t

   (b) Three people take three hours a day to complete a piece of land in three days. If P, h and d represent the number of people, hour and day respectively. Write the law connecting the three variables.

7. (a) A shopkeeper bought $50\ kg$ of sugar at $Tsh\ 2{,}400$ per kg. Due to damage, $5\ kg$ of sugar got spoiled and could not be sold. If the remaining sugar is sold at $Tsh\ 2{,}600$ per kg. Find the percentage profit or loss made by the shopkeeper.

(b) A school is planning to distribute 960 textbooks among three subjects: Mathematics, Science, and English in the ratio $5:7:4$ respectively. Later, the school receives an extra 240 books, which are to be shared among the same subjects in the ratio $2:3:1$. What is the ratio of the total number of books received for Mathematics to Science to English?

8. (a) The terms which are multiples of 4 are deleted from the series of the first one thousand natural numbers. Find the sum of the resulting series.

(b) A ball is dropped from the height of $1458m$ and it rebounds to two third of the height from which if falls. If it continuous to fall and rebound this way. How far will the ball fall down until the ball is coming to rest.

9. (a) A person is standing on top of $15\ meter$-high building A. He observes a base of another building B at an angle of depression of $40°$. At the same time, he sees the top of building B at an angle of elevation of $25°$.

(i) Draw a clear diagram to represent the situation.

(ii) Calculate the height of building B.

(b) A rectangular park has a length of $120\ meters$ and a width of $90\ meters$. A straight path is constructed diagonally from one corner of the park to the opposite corner. If a person walks along the diagonal instead of walking along the two sides, how many meters does he saves?

10. (a) The operation $*$ is defined as $m * n = \begin{cases} 2m - n\ if\ a > b \\ 2n + m\ if\ a < b \end{cases}$.

Find the value of x if $-2 * -1 = x$

(b) John has $Tshs\ 1800$ to spend on oranges. If the price of an orange is decreased by $Tshs\ 20$, he can buy 3 more oranges. Find the price of the orange.

SECTION B: (40 Marks)

Answer **all** questions in this section

11. (a) A circular garden is surrounded by a path that is 1.5 meters wide. The radius of the garden is 7.5 meters. The path is to be paved with bricks. If one square meter requires 12 bricks, how many bricks are needed to pave the path?

(b) The following table shows the distribution of marks (in %) scored by 30 students in mathematics who sat for JIEPSS in a certain school.

Class Interval	$70-80$	$60-70$	$50-60$	$40-50$	$30-40$	$20-30$
Frequency	3	5	y	10	x	2

(i) If half of the majority got below 49% and the other half got above. Find the value of x and y.

(ii) Draw a cumulative frequency curve

12. (a) Given set $A = \{1,2,3\}$ and defines a relation R on $A \to A$ by $(x, y) \in R$.
If $x = y$ and $x + y = 4$.

(i) List all pairs in the relation R

(ii) Draw the relation using arrow diagram.

(b) If $R^{-1} = \{(x, y): y \geq x \text{ and } x \geq 0\}$. Draw the graph of R hence state the domain and range.

13. (a) Find the image of a line $5x + 2y = -5$ after rotation through $90°$ clockwise.

(b) If a point $A(1,2)$ is reflected under the line $y + x = 0$ followed by enlargement by factor $k = 2$ about the point $(-2,3)$. Find the final position of point A

14. (a) Without using formula find the position where the function $f(x)$ turn and the line which divide $f(x)$ curve in to two equal parts. Given that $f(x+1) = 2x^2 + 7x + 6$.

(b) If $h(x) = \begin{cases} \text{a number is squared and then multiplied by } -1. & for \;\; x < 0 \\ \text{a number is doubled and 3 is added to it.} & for \;\; x \geq 0 \end{cases}$

and x represent a number. Sketch the graph of $h(x)$ and state the domain and range.

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